



Enviro Liner 4000

1. Product Description

The Enviro Liner® 4000 series is a specialized polyethylene lining material that combines flexibility, UV stability, and chemical resistance. The Enviro Liner® 4000 series was originally designed as a flexible cap material for landfills. Its good flexibility allows it to accommodate differential settlement that is often seen in landfill applications. This flexibility also allows it to be prefabricated (welded, folded and rolled) so that many jobs can be lined using a one-piece liner. The Enviro Liner® 4000 series is manufactured by Layfield. Each step in the production of your liner panel, from the geomembrane production, to fabrication into a custom panel, is governed by Layfield's registered quality management system. Enviro Liner® 4000 is stabilized against UV degradation with carbon black. If you plan to leave the liner exposed then the additional UV stabilizers that we add to the Enviro Liner® 6000^{HD} series might be of interest to you to maximize the exposed life of the liner.

2. Technical Data

Materials information is on page 2.

3. Installation

Layfield's LLDPE is flexible enough to be prefabricated at our facility into large panels (Up to 27,000 square feet at 30 mil). The prefabricated panel is accordion folded, rolled on a core, and delivered to the job site secured to a pallet. Prefabricated panels can often cover a small project with a single panel. Local labor forces can be used to unroll and unfold the panel, while on larger projects Layfield installation forces can be used to join panels. Layfield has spent years developing innovative thin film seaming technology. All of our primary field welding of LLDPE is based on hot wedge welding technology. Field wedge welding of LLDPE provides strong seams, and fast installations on large projects. Small welds and repairs can be completed with the Enviro Liner welding kit. Ask us about our Enviro Liner® Welding Gun for small repairs.



4. Availability and Cost

Available from Layfield or distributors. Call 425-254-1075 Pacific time, 780-453-6731 Mountain time, or 905-761-9123 Eastern time

5. Manufactured By

Layfield Environmental Systems Corp.
Layfield Geosynthetics & Ind. Fabrics Ltd.

6. Warranty

Products sold will meet Layfield's published specifications. Any extended warranty required by the buyer must be negotiated at the time of order. Extended warranties may be available on this product and may be at extra cost. Full warranty details are available from Layfield.

7. Maintenance

Geomembranes should be inspected at least once per year for damage, stress, or any other detrimental condition. The entire containment area should be visually inspected annually. Layfield provides geomembrane maintenance services on request.

8. Filing Systems

www.LayfieldGroup.com
www.geomembranes.com

9. Enviro Liner 4000 Material Properties

19 Nov 2011	Enviro Liner® 4000 Minimum Properties			
Style	ASTM	Enviro Liner® 4020	Enviro Liner® 4030	Enviro Liner® 4040
Thickness (Minimum Average)	D5199	20 mil 0.5 mm	30 mil 0.75 mm	40 mil 1.0 mm
Thickness (Lowest Individual of 10 values)	D5199	-10%	-10%	-10%
Density (Max)	D792	0.939	0.939	0.939
Tensile Strength at Break	D638 Type IV	76 ppi 13.3 N/mm	114 ppi 20.0 N/mm	152 ppi 26.6 N/mm
Elongation	D638	800%	800%	800%
Tear Resistance	D1004	11 lbs 49 N	16 lbs 71 N	22 lbs 98 N
Puncture Resistance	D4833	32 lbs 142 N	42 lbs 186 N	61 lbs 271 N
Carbon Black Content	D1603	2.0-3.0%	2.0-3.0%	2.0-3.0%
Low Temperature Impact Resistance	D746	-69°F -56°C	-69°F -56°C	-69°F -56°C

10. Shop Seam Strengths

19 Nov 2011	Enviro Liner® 4000 Minimum Shop Seam Strengths			
Style	ASTM	Enviro Liner® 4020	Enviro Liner® 4030	Enviro Liner® 4040
Heat Bonded Seam Strength	D6392 25.4 mm (1") Strip	30 ppi 5.15 N/mm	45 ppi 7.75 N/mm	60 ppi 10.35 N/mm
Heat Bonded Peel Adhesion Strength	D6392 25.4 mm (1") Strip	FTB 25 ppi 4.3 N/mm	FTB 38 ppi 6.53 N/mm	FTB 50 ppi 8.62 N/mm

11. Field Seam Strengths

19 Nov 2011	Enviro Liner® 4000 Minimum Field Seam Strengths			
Style	ASTM	Enviro Liner® 4020	Enviro Liner® 4030	Enviro Liner® 4040
Heat Bonded Seam Strength	D6392 25.4 mm (1") Strip	30 ppi 5.15 N/mm	45 ppi 7.7 N/mm	60 ppi 10.35 N/mm
Heat Bonded Peel Adhesion Strength	D6392 25.4 mm (1") Strip	FTB 25 ppi 4.2 N/mm	FTB 38 ppi 6.5 N/mm	FTB 50 ppi 8.6 N/mm

Layfield
Environmental Systems

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Design | Manufacture | Fabrication | Installation | Maintenance